

# Valvie

Predicting structural valve deterioration

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# Problem

## Core Challenge

There is currently no reliable method to predict the lifespan of a bioprosthetic aortic valve using routine clinical data.

## The Two Essential Questions

- **Durability:** When will the valve fail? (Time-to-event)
- **Severity:** How much has it already deteriorated? (VARC-3 staging)

# 12.6M

People globally living with calcific aortic valve disease  
(Yadgir et al., 2020)

# 14% → 47%

Shift toward bioprosthetic valves between 1997–2014

# 8.2 Years

Median time to SVD in patients  $\leq 50$  years old

# Data



## Dataset Overview

- ✓ **117 Total patients** with unstructured clinical notes (Operative Reports, Progress Notes).
- ✓ **< 10 Confirmed** failure events (8-9 identified).

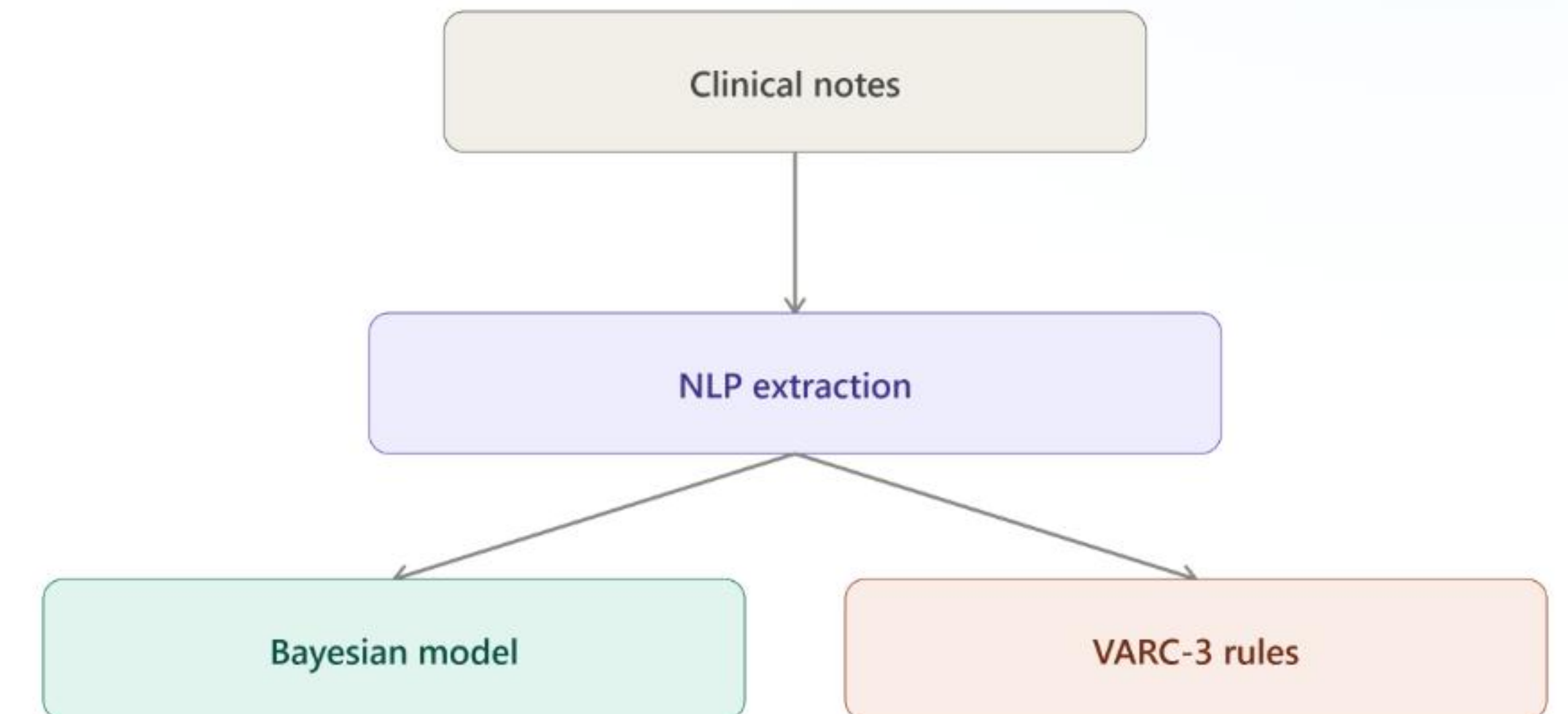


## Machine Learning Challenge

- ✓ **No Explicit Ground Truth:** Absence of ready-made SVD labels or standardized failure timestamps.
- ✓ **Data Sparsity:** Full multimodal overlap (Labs + Meds + Notes) exists for only a small subset of the cohort.

# Our Model

- **Input:** Physicians' clinical notes (free-text format).
- **NLP Extraction:** Automatically extracts useful data (valve type, measurements, medical history) into a structured table.
- **Dual-Model Architecture:**
  - ▶ **Bayesian Model:** Predicts how many years the valve will last.
  - ▶ **Medical Rules (VARC-3):** Determines the current stage of valve deterioration.



# Innovation 1: Learning from Minimal Data

## Literature-Informed Priors

Utilizing existing literature (Capodanno 2017, Trimaille 2025) instead of an unrealistic attempt to train from scratch.

## Full Traceability

Every prediction is completely transparent and explainable. The model pinpoints **exactly** which clinical factors and specific text excerpts led to the estimation.

### SVD Stage 0

- Yearly clinical and echocardiographic follow-up

### SVD Stage 1

- Repeat echocardiography 3-6 months after initial diagnosis (then if stable every 12 months)
- Trial of anticoagulation if leaflet thickening is suspected

### SVD Stage 2S

- Repeat echocardiography 3-6 months after initial diagnosis (then if stable every 6-12 months)

### SVD Stage 2R

- Repeat echocardiography 3-6 months after initial diagnosis (then if stable every 6-12 months)

### SVD Stage 2RS

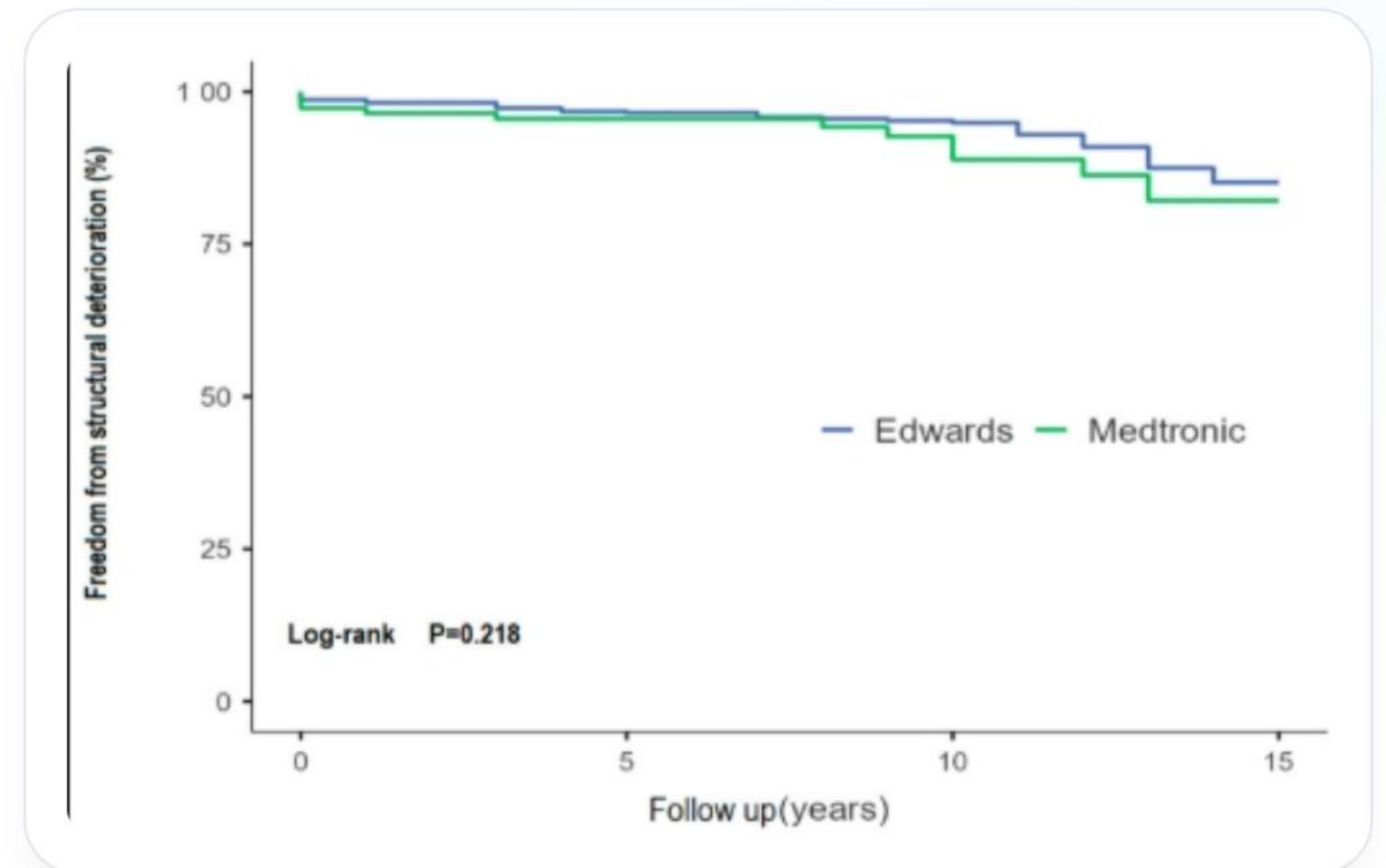
- Clinical evaluation every 3-6 months
- Repeat echocardiography 3-6 months after initial diagnosis (then if stable every 6-12 months)
- Consider intervention if symptomatic

### SVD Stage 3

- Consider intervention
- If asymptomatic with preserved LVEF consider clinical evaluation every 3-6 months and repeat echocardiography every 6 months

## Innovation 2: More Data, No New Patients

- We developed an algorithm that reconstructs individual patient data from published scientific survival curves.
- We identified and utilized an open-source study of 918 patients specifically for SVD (Hlavička 2026).
- This allows the model to learn from hundreds of additional cases, without requiring any new patient data collection.



# Clinician's Contribution



## Medical Guidance

Comprehensive explanation of the clinical background (SVD mechanisms, physiology) to the technical team, ensuring medical accuracy in our design.



## Feature Selection

Targeted guidance in selecting the most critical clinical variables that feed into the prediction model.



## Data Annotation

Specialized annotation of unstructured clinical notes, primarily focusing on confirming the presence of SVD and classifying its severity.

# Results

- > **100% Detection Rate:** Successfully identified all 8 confirmed SVD cases in the original dataset (117 patients) — zero missed events.
- > **Reintervention Detector:** Achieved 100% Recall and 80% Precision (F1-Score = 0.889).



## Case Study

### Patient\_035

Biocor (2010) → ViV-TAVR (2024)

**Prediction: 15.2 yrs** [80% CI: 13.1–19.3]

**Actual Failure: 14 yrs**

# Thank You

Questions & Discussion

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